# Functional & Performance Testing Template

## Model Performance Test

| | |
|---|---|
| Date | 28 February 2026 |
| Team ID | LTVIP2026TMIDS66052 |
| Project Name | IntelliSQL: Intelligent SQL Querying with LLMs Using Gemini Pro |
| Maximum Marks | |

**Test Scenarios & Results**

| Test Case ID | Scenario (What to test) | Test Steps (How to test) | Expected Result | Actual Result | Pass/Fail |
|---|---|---|---|---|---|
| **FT-01** | **Text Input Validation** (Query Phrasing) | Enter valid English queries (e.g., "Show marks") and invalid gibberish. | Valid queries generate SQL; invalid inputs are handled without crashes. | Successfully filtered valid/invalid text. | Pass |
| **FT-02** | **Schema Parameter Validation** (Specific Columns) | Query specific student names or marks within and outside database ranges. | Accepts valid column targets (NAME, MARKS); ignores non-existent table requests. | Correctly mapped query to table schema. | Pass |
| **FT-03** | **SQL Content Generation** (LLM Logic) | Provide complete English prompts and click **"Get Answer"**. | Correct SQL is generated and cleaned by the Regex layer. | Generated clean SELECT statements. | Pass |
| **FT-04** | **API Connection Check** (Gemini Authentication) | Verify if the GOOGLE_API_KEY is loaded and the model responds. | API responds successfully and initializes the session. | 200 OK / Successful handshake. | Pass |
| **PT-01** | **Response Time Test** (Latency) | Use a timer to check the time from clicking "Get Answer" to result display. | Content generation and DB fetch should be under 3 seconds. | **Avg: 1.8 seconds** (using Gemini Flash). | Pass |
| **PT-02** | **API Speed Test** | Simulate multiple | The API should | Stable throughput | Pass |

| | | | | | |
|---|---|---|---|---|---|
| | (Concurrency) | rapid-fire query calls to the Gemini model. | process requests without significant slowdown or throttling. | with no degradation. | |
| **PT-03** | **Database Load Test** (SQLite Handling) | Run queries that require scanning all rows in the STUDENTS table. | Results should render smoothly without freezing the Streamlit UI. | Rapid rendering of table dataframes. | Pass |